Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** A set is a collection of well-defined objects or elements.

____ **2.** A set which does not contain any element is called a void (or empty) set.

____ **3.** A set which consists of a definite (countable) number of elements is called a finite set.

____ **4.** Two sets A and B are said to be equal if and only if they have exactly the same or identical elements.

____ **5.** If $A \subseteq B$ and $A \neq B$, then A is called the proper subset of set B.

____ **6.** Two sets A and B are disjoint if $A \cap B = \emptyset$.

____ **7.** A fraction a/b is said to be in simplest (lowest) form if a and b are relatively prime or GCF(a, b) = 1.

____ **8.** Every natural number can be expressed as a product of primes and this factorization is unique.

____ **9.** If a is a rational number and b is an irrational number, then ab is always a rational number.

____ **10.** For two positive integers a and b in the division lemma (a = bq + r), we say a is divisible by b if the remainder r is zero.

____ **11.** If a is a natural number then it is either prime or composite.

II. Choose the best answer among the given alternatives.

____ **12.** Given universal set U = {1, 2, 3, 4, 5, 6, 7, 8, 9}, A = {2, 5, 7, 9}, and B = {2, 3, 5, 7, 8}. Find $A' \cap B'$.

A. { } **B.** {1, 4, 6} **C.** {1, 3, 8, 9} **D.** {2, 5, 7}

____ **13.** Find GCF(12, 18) × LCM(12, 18).

A. 216 **B.** 200 **C.** 36 **D.** 108

____ **14.** In a bookstore there are 125 different books to be distributed to 9 students. If the store shares these books equally, how many books will each student receive and how many books will be left?

A. 8 & 13 **B.** 9 & 13 **C.** 8 & 9 **D.** 13 & 8

____ **15.** Which of the following are well-defined sets?

A: Collection of students in your class
B: Collection of beautiful girls in your class
C: Collection of consonants of the English alphabet
D: The collection of kind students in your school

A. A only
B. B & D
C. C only
D. A & C
E. All of them

____ **16.** Suppose that A is the set of positive even numbers. Which of the following is true?

A. $4 \in A$ **B.** $5 \in A$ **C.** $-2 \in A$ **D.** $0 \in A$

____ **17.** Which of the following is an infinite set?

A. $A = \{x \mid x \in \mathbb{R} \text{ and } 5 < x < 6\}$
B. $B = \{0\}$
C. C is the set of odd natural numbers divisible by 2
D. D = { }

____ **18.** Which of the following statements is false?

A. Any set is a subset of itself
B. Empty set is a subset of every set
C. For any set A, A is not a proper subset of itself
D. The number of proper subsets of set A with n elements is 2n - 1

____ **19.** Which of the following set identities is false?

A. $(A \cup B)' = A' \cap B'$ **B.** $(A \cap B)' = A' \cup B'$
C. $\emptyset \cup A = \emptyset$ **D.** $U \cap A = A$

____ **20.** For any two sets A and B, which of the following is false?

A. $(A')' = A$ **B.** $A \subseteq B \Leftrightarrow B' \subseteq A'$

C. $A - B = A \cap B'$ **D.** $A \Delta B = (A \setminus B) \cap (B \setminus A)$

____ **21.** Given A = {0, 2, 3, 4, 5} and B = {0, 1, 2, 3, 5, 7, 9}. Find the symmetric difference A Δ B.

A. {0, 1, 3} **B.** {1, 4, 7, 9} **C.** empty set **D.** {0, 2, 3, 5}

____ **22.** If the number of elements of set A is 7 and the number of elements of set B is 4, then the number of elements in Cartesian product A × B is:

A. 24 **B.** 25 **C.** 28 **D.** 11

____ **23.** For any two sets A and B, which of the following is false?

A. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ **B.** $n(A') = n(U) - n(A)$

C. $n(A \setminus B) = n(A) - n(B)$ **D.** $n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$

____ **24.** Consider universal set U = {1, 2, 3 ... 14}, A = {2, 3, 5, 7, 11}, and B = {2, 4, 8, 9, 10, 11}. Which statement is true?

A. $(A \cup B)' = \{1, 4, 6, 12, 13, 14\}$ **B.** $A \cap B = A' \cup B'$

C. $A \Delta B = (A \cap B)'$ **D.** $A \setminus B = \{3, 5, 7\}$

____ **25.** Find the greatest common factor (GCF) of 180, 270, and 1,080 using prime factorization.

A. 30 **B.** 45 **C.** 90 **D.** 180

____ **26.** Given that 77,9x2 is a 5-digit number where x is the tens digit. If this number is divisible by 6, what are all possible values of x?

A. 1, 4, 7 **B.** 2, 5, 8 **C.** 0, 3, 6, 9 **D.** 2, 4, 6